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✡ Except for Thyroid hormones, which are lipids-soluble, all amino acid-derived hormones bind to cell membranes receptors that are located
↳ at least in part, on the ~~extracellll~~ ~~extracat~~ extracellular surface of the cell membrane.

They do not directly affect the transcription of target genes → ~~I~~nstead initiate a signaling cascade that is carried out by a molecule called a Second Messenger

~~The~~ ~~Sec~~

The second messenger used by most hormones is cyclic adenosine ~~mon~~ monophosphate

cAMP ← make sure to remember this.

cAMP 2nd msgr sys, a water ~~soluble~~ soluble hormone binds to its receptor in the cell membrane
AKA G-protein the binding of a hormone activates the G-protein component

The ~~G~~ activated G-protein in turn activates-
✡ an enzyme → Adenylyl cyclase

aka adenylate cyclase

converts to ATP